Indirect Space Charge Maths

1 Bassetti-Erskine Formula

The Bassetti-Erskine formula [1] for the electric field is given by:

$$\mathcal{E}(z) = E_x + iE_y = i\sqrt{\frac{2\pi}{\sigma_x^2 - \sigma_y^2}}\left[e^{-S^2/2}w\left(\frac{\bar{\omega}}{\sqrt{2(\sigma_x^2 - \sigma_y^2)}}\right) - w\left(\frac{\bar{z}}{\sqrt{2(\sigma_x^2 - \sigma_y^2)}}\right)\right]$$

with $z = x + iy$, $\omega = \frac{\sigma_y}{\sigma_x}x + i\frac{\sigma_x}{\sigma_y}y$ and $S^2 = \frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2}$. $w(z)$ is the Fadeeva function related to the complex error function.

2 Circular Chamber

This section is inspired by work by Miguel Furman [2].

In 2D (using cgs units) an expression for the electric field from a source $\rho(\mathbf{x})$ is given using a Green's function

$$E(\mathbf{x}) = 2\int d^2\mathbf{x}' \frac{\mathbf{x} - \mathbf{x}'}{|\mathbf{x}' - \mathbf{x}|^2}\rho(\mathbf{x}') \qquad \mathbf{G}(\mathbf{x}, \mathbf{x}') = 2\frac{\mathbf{x} - \mathbf{x}'}{|\mathbf{x}' - \mathbf{x}|^2}$$

Using the complex electric field $\mathcal{E}(z) = E_x + iE_y$ and complex position coordinate $z = x + iy$, we can express this as:

$$\mathcal{E}(z) = E_x + iE_y = 2\int d^2z' \frac{\rho(z')}{\bar{z} - \bar{z}'} \qquad \mathcal{G}(z, z') = \frac{2}{\bar{z} - \bar{z}'}$$

A point charge at position $\mathbf{x}$ inside a circle of radius R (centred on the origin) produces a negatively charged image charge at the position $\mathbf{x}_i = \frac{R^2}{|\mathbf{x}|^2}\mathbf{x}$ or in the complex formulation $z_i = \frac{R^2}{\bar{z}}$

Using this we can construct a new Green's function for the image charge:

$$\mathcal{G}_i(z, z') = -\frac{2}{\bar{z} - \bar{z}_i'}$$

It can be shown

$$\frac{1}{\bar{z}-\bar{z}_i'} = \frac{1-\bar{z}_i'/\bar{z}+\bar{z}_i'/\bar{z}}{\bar{z}(1-\bar{z}_i'/\bar{z})} = \frac{1}{\bar{z}} + \frac{\bar{z}_i'/\bar{z}}{\bar{z}-\bar{z}_i'} = \frac{1}{\bar{z}} + \frac{z_i/\bar{z}}{\bar{z}z_i/\bar{z}_i'-z_i} = \frac{1}{\bar{z}} - \frac{z_i/\bar{z}}{z_i-z'}$$

$$\mathcal{G}_i(z,z') = \frac{-2}{\bar{z}} + \frac{z_i/\bar{z}}{z_i-z'}$$

Substituting this back into our convolution integral:

$$\mathcal{E}_i(z) = \int d^2z' \left(\frac{-2}{\bar{z}} + \frac{z_i/\bar{z}}{z_i-z'}\right)\rho(z') = \frac{-2}{\bar{z}}Q + \frac{z_i}{\bar{z}}2\int d^2z' \frac{\rho(z')}{z_i-z'}$$

Where $Q = 1$ is the normalised total charge. The remaining integral on the right is simply the direct field formula except $\bar{z}$ and $\bar{z}'$ are replaced with z_i and z'. This is equivalent direct field evaluated at $\bar{z}_i$

$$\mathcal{E}_i(z) = \frac{-2}{\bar{z}} + \frac{z_i}{\bar{z}}\mathcal{E}_d(\bar{z}_i)$$

3 Rectangular Chamber

Poisson's equation ($\nabla^2\phi = -4\pi\rho$) for a Gaussian source inside a PEC rectangle may be solved using a Fourier series.Consider a Gaussian source at position (x_0, y_0), with widths σ_x, σ_y inside a PEC rectangle with lengths L_x, L_y with its bottom left corner at the origin.

We first express the potential $\phi(x,y)$ and source $\rho(x,y)$ as sin Fourier series since this guarantees $\phi = 0$ at the boundary (the PEC boundary condition).

$$\phi(x,y) = \sum_{n,m}\phi_{nm}\sin(k_nx)sin(k_my) \qquad \rho(x,y) = \sum_{n,m}\rho_{nm}\sin(k_nx)sin(k_my)$$

Poisson's equation can then be solved in k-space

$$\nabla^2\phi(x,y) = -\sum_{n,m}\phi_{nm}(k_n^2+k_m^2)\sin(k_nx)sin(k_my)$$

$$\phi_{nm} = \frac{4\pi\rho_{nm}}{(k_n^2+k_m^2)}$$

Our source has the form:

$$\rho(x,y) = \frac{1}{2\pi\sigma_x\sigma_y}exp\left(-\frac{(x-x_0)^2}{2\sigma_x^2} - \frac{(y-y_0)^2}{2\sigma_y^2}\right)$$

$$\rho_{nm} = \frac{4}{L_xL_y}\iint \rho(x,y)\sin(k_nx)sin(k_my)dxdy := I_xI_y$$

where I_x are integrals over $\mathbb{R}$. Which is an approximation that the entire source is within the rectangle.

$$I_x = \frac{2}{L_x\sqrt{2\pi}\sigma_x}\int dxdy \exp\left[-\frac{(x-x_0)^2}{2\sigma_x^2}\right]\sin(k_n x)$$

$$= \frac{1}{iL_x\sqrt{2\pi}\sigma_x}\int dxdy/exp\left[-\frac{(x-x_0)^2}{2\sigma_x^2}\right]e^{ik_n x} - c.c.$$

$$= \frac{1}{iL_x\sqrt{2\pi}\sigma_x}\int dxdy \exp\left[-\frac{x^2}{2\sigma_x^2} + \left(\frac{x_0}{\sigma_x^2} + ik_n x\right)x - \frac{x_0^2}{2\sigma_x^2}\right] - c.c.$$

$$= \frac{1}{iL_x\sqrt{2\pi}\sigma_x}\exp\left[-\frac{x_0^2}{2\sigma_x^2}\right]\int dxdy \exp\left[-\frac{x^2}{2\sigma_x^2} + \left(\frac{x_0}{\sigma_x^2} + ik_n x\right)x\right] - c.c.$$

Using known formula for Gaussian integrals

$$= \frac{1}{iL_x}\exp\left[ik_n x_0 - \frac{1}{2}k_n^2\sigma_x^2\right] - c.c.$$

$$= \frac{2}{L_x}\exp\left[-\frac{1}{2}k_n^2\sigma_x^2\right]\sin\left(k_n x_0\right)$$

Therefore,

$$\rho_{nm} = \frac{4}{L_x L_y}\exp\left[-\frac{1}{2}(k_n^2\sigma_x^2 + k_m^2\sigma_y^2)\right]\sin\left(k_n x_0\right)\sin\left(k_m y_0\right)$$

Thus, altogether:

$$\phi_{nm} = \frac{16\pi}{L_x L_y}\frac{\exp\left[-\frac{1}{2}(k_n^2\sigma_x^2 + k_m^2\sigma_y^2)\right]}{k_n^2 + k_m^2}\sin\left(k_n x_0\right)\sin\left(k_m y_0\right)$$

Electric fields are expressed as gradients of the potential $\mathbf{E} = -\nabla\phi$

$$E_x(x,y) = -\sum_{n,m} k_n\phi_{nm}\cos(k_n x)sin(k_m y)$$

$$E_y(x,y) = -\sum_{n,m} k_m\phi_{nm}\sin(k_n x)cos(k_m y)$$

References

[1] M Bassetti and G A Erskine. "CLOSED EXPRESSION FOR THE ELECTRICAL FIELD OF A TWO-DIMENSIONAL GAUSSIAN CHARGE". In: ().

[2] Miguel A. Furman. "Electric field of a 2D elliptical charge distribution inside a cylindrical conductor". In: *Physical Review Special Topics - Accelerators and Beams* 10.8 (Aug. 7, 2007), p. 081001. ISSN: 1098-4402. DOI: 10.1103/PhysRevSTAB.10.081001. URL: https://link.aps.org/doi/10.1103/PhysRevSTAB.10.081001 (visited on 10/22/2026).